

5. The symmetric group

The *symmetric group* S_n is the group of all permutations of $X = \{1, \dots, n\}$ under composition, that is, S_X of Definition 4.2. It makes the ideas of Chapter 4 concrete. Its elements move labels, its cycle decomposition describes orbits, and conjugation changes the labelling. In physics, labels can mark particles, identical degrees of freedom, or the positions of indices in a tensor. We first develop the rules for manipulating permutations, then use them to organise motions and arrays of physical quantities.

5.1. PERMUTATIONS, CYCLES, AND CYCLE DECOMPOSITION

READING. [Moore, §6.2, pp. 59–61].

A permutation is written in two rows, each image below its argument,

$$\varphi = \begin{pmatrix} 1 & 2 & \cdots & n \\ \varphi(1) & \varphi(2) & \cdots & \varphi(n) \end{pmatrix},$$

and permutations are composed *right to left*, $(\varphi\psi)(i) = \varphi(\psi(i))$.

Definition 5.1 (Cycle). Let $i_1, \dots, i_r$ be distinct elements of X . The permutation carrying

$$i_1 \mapsto i_2 \mapsto \cdots \mapsto i_r \mapsto i_1$$

and fixing every other element is an *r-cycle*, written $(i_1 i_2 \cdots i_r)$. A 2-cycle is a *transposition*. Two cycles are *disjoint* if no element appears in both.

cycle
轮换

transposition
对换

The notation is not unique. A cycle may be started anywhere,

$$(2\ 3\ 4) = (3\ 4\ 2) = (4\ 2\ 3),$$

and reversing it gives the inverse, $(2\ 3\ 4)^{-1} = (4\ 3\ 2)$. Disjoint cycles commute, since each fixes what the other moves,

$$(2\ 3\ 4)(5\ 6) = (5\ 6)(2\ 3\ 4),$$

while cycles sharing an element need not,

$$(1\ 2)(2\ 3) = (1\ 2\ 3), \quad (2\ 3)(1\ 2) = (1\ 3\ 2).$$

cycle decomposition
轮换分解

Theorem 5.2 (Cycle decomposition). Every $\varphi \in S_n$ has a complete factorization into disjoint cycles, one for each orbit on X of $\langle \varphi \rangle = \{\varphi^k : k \in \mathbb{Z}\}$, including a 1-cycle for each fixed point. It is unique up to the order of the cycles and cyclic rotation of their entries.

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Proof. One orbit. For $x \in X$, let r be the first step at which the sequence $x, \varphi(x), \varphi^2(x), \dots$ repeats an entry. Since X has n elements, $1 \leq r \leq n$, and the r points $x, \varphi(x), \dots, \varphi^{r-1}(x)$ are distinct. Write $\varphi^r(x) = \varphi^i(x)$ with $0 \leq i < r$. Applying φ^{-i} gives

$$\varphi^{r-i}(x) = x.$$

If $i > 0$ this repeats x at step $r - i < r$, against the choice of r . So $i = 0$ and the sequence closes into a cycle,

$$x \mapsto \varphi(x) \mapsto \dots \mapsto \varphi^{r-1}(x) \mapsto x.$$

Both φ and φ^{-1} move along this chain, so its entries are exactly the orbit of x under $\langle \varphi \rangle$.

All orbits. The orbits partition X by Proposition 4.9. Form the cycle above for each orbit, including (x) when x is fixed. These cycles are disjoint, and their product agrees with φ at every point of X .

Uniqueness. In any complete factorization into disjoint cycles, the cycle containing x must follow $x, \varphi(x), \varphi^2(x), \dots$ until it returns to x , since every other factor fixes each entry of that cycle. It is therefore the cycle just constructed, and only its starting entry and the order of the disjoint cycles can vary. $\square$

The orbits of size one are the fixed points, and a cycle notation usually leaves them out. Writing them in gives the *complete factorization*, one cycle for each orbit,

$$\varphi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 3 & 2 & 6 & 5 & 4 \end{pmatrix} = (1)(2\ 3)(4\ 6)(5),$$

which has four cycles where the usual notation $(2\ 3)(4\ 6)$ shows two.

Proposition 5.3. *The order of a permutation is the least common multiple of the lengths of the cycles in its complete factorization.*

Proof. Let $\varphi = \gamma_1 \cdots \gamma_t$ be disjoint, of lengths $r_1, \dots, r_t$. Disjoint cycles commute, so $\varphi^m = \gamma_1^m \cdots \gamma_t^m$, and these factors move disjoint sets of points. So φ^m is the identity exactly when every γ_i^m is, that is, when r_i divides m for each i . $\square$

Example 5.4 (Reading off a decomposition). For

$$\varphi = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix},$$

follow $1 \mapsto 2 \mapsto 4 \mapsto 3 \mapsto 1$, so $\varphi = (1\ 2\ 4\ 3)$, of order 4. For $\psi = (1\ 2\ 3)(4\ 5) \in S_6$ the complete factorization is $(1\ 2\ 3)(4\ 5)(6)$ and the order is $\text{lcm}(3, 2, 1) = 6$.

Quiz 5.1. Write the permutation $1 \mapsto 3, 2 \mapsto 5, 3 \mapsto 6, 4 \mapsto 2, 5 \mapsto 4, 6 \mapsto 1$ of S_6 in cycles, and give its order. How many elements of S_4 have order 4?

5.2. CAYLEY'S THEOREM

READING. [Moore, §6.1, pp. 58–59].

Theorem 5.5 (Cayley). *Every group G is isomorphic to a subgroup of S_G . If $|G| = n$, then G is isomorphic to a subgroup of S_n .*

Cayley's theorem
凯莱定理

Proof. Let G act on itself by left multiplication, $h \cdot g = hg$, as in Example 4.4. The action is a homomorphism

$$L : G \rightarrow S_G, \quad L(h)(g) = hg,$$

and $L(h) = \text{id}$ forces $h = he = e$, so L is injective by Proposition 3.3. Therefore $G \cong \text{im } L$, a subgroup of S_G . Numbering the elements of G as $1, \dots, n$ identifies S_G with S_n . $\square$

To make the numbering explicit, let $f : G \rightarrow \{1, \dots, n\}$ be a bijection. Left multiplication, written in these labels, is the permutation

$$L_f(h) = f \circ L(h) \circ f^{-1} \in S_n.$$

Suppose we choose another numbering f' . The permutation $\sigma = f' \circ f^{-1}$ converts an old label into its new label, so $f' = \sigma \circ f$. Therefore

$$\begin{aligned} L_{f'}(h) &= f' \circ L(h) \circ (f')^{-1} \\ &= \sigma \circ f \circ L(h) \circ f^{-1} \circ \sigma^{-1} \\ &= \sigma L_f(h) \sigma^{-1}. \end{aligned}$$

Reading the last expression right to left, we convert a new label back to the old label, apply the old permutation, and convert the result to its new label. Consequently the two subgroups of S_n satisfy

$$\text{im } L_{f'} = \sigma(\text{im } L_f)\sigma^{-1}.$$

Left multiplication defines $\text{im } L \subset S_G$ intrinsically. A numbering realizes this subgroup inside S_n ; different numberings give conjugate subgroups, each isomorphic to the same group G .

Example 5.6 (Cyclic groups). For $G = \mathbb{Z}_n$, label the elements $0, 1, \dots, n-1$ by $1, 2, \dots, n$, respectively. The element 1 acts by adding one to everything, which is the n -cycle $(1\ 2\ \dots\ n)$. So $\mathbb{Z}_n \cong \langle (1\ 2\ \dots\ n) \rangle$, and for $n = 3$,

$$\mathbb{Z}_3 \cong \{e, (1\ 2\ 3), (1\ 3\ 2)\} \subset S_3.$$

For a concrete change of numbering, take $G = \mathbb{Z}_4$ and choose

element of $\mathbb{Z}_4$	0	1	2	3
old label f	1	2	3	4
new label f'	1	3	2	4

The relabelling is $\sigma = (2\ 3)$. Addition by 1 gives $L_f(1) = (1\ 2\ 3\ 4)$ in the old labels and $L_{f'}(1) = (1\ 3\ 2\ 4)$ in the new ones. They satisfy

$$(2\ 3)(1\ 2\ 3\ 4)(2\ 3)^{-1} = (1\ 3\ 2\ 4).$$

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Thus the image changes from $\langle(1\ 2\ 3\ 4)\rangle$ to $\langle(1\ 3\ 2\ 4)\rangle$. These subgroups are distinct. Their unique elements of order two are respectively $(1\ 3)(2\ 4)$ and $(1\ 2)(3\ 4)$. Both still describe the same addition modulo four.

Example 5.7 (The Klein four-group). Write $V = \{e, a, b, c\}$ with $a^2 = b^2 = c^2 = e$ and $ab = c$. The numbering and the labels obtained by left multiplication are

x	e	a	b	c
$f(x)$	1	2	3	4
$f(ax)$	2	1	4	3
$f(bx)$	3	4	1	2
$f(cx)$	4	3	2	1

Reading each of the last three rows as a permutation of the labels gives

$$L_f(a) = (1\ 2)(3\ 4),$$

$$L_f(b) = (1\ 3)(2\ 4),$$

$$L_f(c) = (1\ 4)(2\ 3).$$

These three, with the identity, are a subgroup of S_4 isomorphic to V . Each row of the multiplication table of G is a permutation of the elements, and Cayley's theorem is the observation that these permutations multiply as the group does.

Example 5.8 (A smaller home for D_4). Cayley's theorem puts D_4 , of order 8, inside S_8 . Its action on the four vertices of the square does better. With the vertices numbered around the square as in Example 4.7, the quarter turn and the reflection in the horizontal axis are

$$A = (1\ 2\ 3\ 4), \quad B = (1\ 4)(2\ 3), \quad AB = (2\ 4),$$

so D_4 is isomorphic to a subgroup of S_4 . The embedding of Cayley's proof is the regular action, which is free, while the vertex action is not, and a faithful action on a smaller set gives a smaller symmetric group.

Quiz 5.2. Write out the eight permutations of $\{1, 2, 3, 4\}$ given by D_4 acting on the vertices, and check that they are closed under composition. Which elements of S_4 are missing, and what do they do to the square?

5.3. TRANSPOSITIONS, PARITY, AND THE ALTERNATING GROUP

READING. [Moore, §6.3, pp. 61–65].

Proposition 5.9. Every r -cycle with $r \geq 2$ is a product of transpositions,

$$(a_1\ a_2\ \dots\ a_r) = (a_1\ a_r)(a_1\ a_{r-1}) \cdots (a_1\ a_2),$$

so the transpositions generate S_n for $n \geq 2$. A 1-cycle is the identity, the empty product.

Proof. Read the right-hand side from the right. It sends a_1 to a_2 , and it sends a_i for $i \geq 2$ first to a_1 and then to a_{i+1} , with a_r going to a_1 . Points outside the cycle are untouched. Every permutation is a product of cycles by Theorem 5.2. $\square$

The adjacent transpositions $s_i = (i \ i+1)$ generate S_n . To build an arbitrary exchange $(a \ b)$, first suppose $a + 1 < b$ and conjugate it by s_a . Reading $s_a(a \ b)s_a$ from right to left gives

$$\begin{aligned} a + 1 &\mapsto a \mapsto b \mapsto b, \\ b &\mapsto b \mapsto a \mapsto a + 1. \end{aligned}$$

The label a returns to itself, and every other label is fixed. Thus conjugation has changed the exchange of a and b into the exchange of $a + 1$ and b . Since $s_a^2 = e$, we can also reverse this step.

$$s_a(a \ b)s_a = (a+1 \ b), \quad (a \ b) = s_a(a+1 \ b)s_a.$$

Repeating this step moves the first entry towards b , until the exchange is between the adjacent labels $b - 1$ and b . For example, in S_5 ,

$$\begin{aligned} (1 \ 5) &= s_1(2 \ 5)s_1 \\ &= s_1s_2(3 \ 5)s_2s_1 \\ &= s_1s_2s_3(4 \ 5)s_3s_2s_1 \\ &= s_1s_2s_3s_4s_3s_2s_1. \end{aligned}$$

The general expression is therefore, for $b \geq a + 2$,

$$(a \ b) = s_a s_{a+1} \cdots s_{b-2} s_{b-1} s_{b-2} \cdots s_{a+1} s_a.$$

For $b = a + 1$ the exchange is already s_a . Since every permutation is a product of transpositions, these adjacent exchanges generate S_n .

For $n \geq 2$, let $c = (1 \ 2 \ \cdots \ n)$. Conjugating by c shifts both entries of an adjacent exchange forward by one. The exchange $(1 \ 2)$ becomes $(2 \ 3)$, then $(3 \ 4)$, and so on up to $(n - 1 \ n)$. Starting from s_1 , after $i - 1$ such shifts we obtain s_i , so

$$c^{i-1} s_1 c^{-(i-1)} = s_i \quad (1 \leq i < n).$$

Thus s_1 and c produce every adjacent transposition, and $S_n = \langle s_1, c \rangle$.

A presentation of S_n , stated here without proof, is

$$S_n = \langle s_1, \dots, s_{n-1} \mid s_i^2 = 1, \quad s_i s_j = s_j s_i \ (|i - j| \geq 2), \quad s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1} \rangle.$$

Each s_i squares to the identity, disjoint exchanges commute, and both sides of the last relation exchange i with $i + 2$.

A transposition pair $\tau\tau = e$ can be inserted into any product, increasing the number of factors by two without changing the permutation. The distinction between an even and an odd number of factors is the *parity* of a permutation.

Definition 5.10 (Even and odd). A permutation is *even* if it can be written as a product of an even number of transpositions, and *odd* if it can be written as a product of an odd number of transpositions. The identity is even, as the empty product.

parity
奇偶性
even permutation
偶置换
odd permutation
奇置换

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The determinant of a permutation matrix shows that every decomposition of a given permutation has the same parity.

sign
符号

Definition 5.11 (Permutation matrix and sign). For $\varphi \in S_n$ let $P(\varphi)$ be the matrix with $P(\varphi)\hat{e}_i = \hat{e}_{\varphi(i)}$, that is $P(\varphi)_{ji} = \delta_{j,\varphi(i)}$, and set

$$\text{sgn}(\varphi) := \det P(\varphi).$$

Proposition 5.12 (Parity and sign). $P : S_n \rightarrow GL(n, \kappa)$ is an injective homomorphism, and $\text{sgn} : S_n \rightarrow \{\pm 1\}$ is a homomorphism with $\text{sgn}(\tau) = -1$ for every transposition τ . If φ is a product of k transpositions then $\text{sgn}(\varphi) = (-1)^k$, so the parity of k depends only on φ .

Proof. $P(\varphi)P(\psi)\hat{e}_i = P(\varphi)\hat{e}_{\psi(i)} = \hat{e}_{\varphi\psi(i)}$, so P is a homomorphism, and $P(\varphi) = \mathbb{1}$ forces $\varphi = \text{id}$. The determinant is multiplicative, so sgn is a homomorphism, and its values are ± 1 because $P(\varphi)^{-1} = P(\varphi^{-1})$ has integer entries. The matrix of a transposition is $\mathbb{1}$ with two columns exchanged, of determinant -1 . If two decompositions of φ contain k and ℓ transpositions, respectively, their determinants give

$$(-1)^k = \det P(\varphi) = (-1)^\ell.$$

Hence k and ℓ have the same parity. Every permutation is therefore either even or odd, and $\text{sgn}(\varphi)$ is $+1$ for an even permutation and -1 for an odd one. $\square$

Proposition 5.13. An r -cycle has sign $(-1)^{r-1}$. If the complete factorization of $\varphi \in S_n$ has t cycles then $\text{sgn}(\varphi) = (-1)^{n-t}$.

Proof. An r -cycle is a product of $r - 1$ transpositions by Proposition 5.9. If the complete factorization has cycles of lengths $r_1, \dots, r_t$ then $\sum_i r_i = n$, since every point lies in exactly one cycle, and $\text{sgn}(\varphi) = \prod_i (-1)^{r_i-1} = (-1)^{n-t}$. $\square$

Levi-Civita symbol
列维-奇维塔符号

Example 5.14 (The sign in vector calculus). The *Levi-Civita symbol* packages the sign of a permutation into indices. For $\sigma \in S_n$, set

$$\epsilon_{\sigma(1)\dots\sigma(n)} = \text{sgn}(\sigma),$$

and set it to zero when an index is repeated. Thus $\epsilon_{123} = \epsilon_{231} = 1$ and $\epsilon_{132} = -1$. In three dimensions,

$$(\vec{a} \times \vec{b})_i = \sum_{j,k=1}^3 \epsilon_{ijk} a_j b_k.$$

Exchanging j and k changes the sign, giving $\vec{b} \times \vec{a} = -\vec{a} \times \vec{b}$. The same permutation sign appears in the determinant expansion,

$$\det M = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n M_{i,\sigma(i)}.$$

alternating group
交错群

Parity therefore already occurs in familiar calculations with areas, volumes, and cross products.

Definition 5.15 (Alternating group). $A_n := \ker(\text{sgn})$, the even permutations in S_n .

It is a subgroup, being a kernel. For $n \geq 2$, the odd permutations form the coset $(1\ 2)A_n$. Multiplication by $(1\ 2)$ reverses parity. For any odd σ , $(1\ 2)\sigma$ is even, so $\sigma = (1\ 2)((1\ 2)\sigma) \in (1\ 2)A_n$. Thus every permutation belongs to one of the two cosets,

$$S_n = A_n \sqcup (1\ 2)A_n.$$

There are two cosets, so $[S_n : A_n] = 2$ and

$$|A_n| = \frac{n!}{2}.$$

Example 5.16 (Small alternating groups). $A_2 = \{e\}$ and $A_3 = \{e, (1\ 2\ 3), (1\ 3\ 2)\} \cong \mathbb{Z}_3$. Next,

$$A_4 = \{e\} \cup \{\text{the eight 3-cycles}\} \cup \{(1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\},$$

of order 12. The three double transpositions with e form the Klein four-group of Example 5.7. While A_3 is abelian, A_4 is not,

$$(1\ 2\ 3)(1\ 2\ 4) = (1\ 3)(2\ 4), \quad (1\ 2\ 4)(1\ 2\ 3) = (1\ 4)(2\ 3).$$

Proposition 5.17. A_4 has no subgroup of order 6.

Proof. Suppose $H \subset A_4$ is a subgroup with $|H| = 6$, so $[A_4 : H] = 2$ and the two cosets H and gH partition A_4 for any $g \notin H$. Then $g^2 \in H$, since $g^2 \in gH$ would give $g \in H$. So every square in A_4 lies in H . A 3-cycle σ satisfies $\sigma^3 = e$, hence $\sigma = \sigma^4 = (\sigma^2)^2$, and is a square. All eight 3-cycles would therefore lie in H , which has six elements. $\square$

This is the counterexample promised in §4.5. A divisor of the order of a group need not be the order of a subgroup.

Remark 5.18 (Sylow's first theorem). If $|G| = p^a m$ with p prime and p not dividing m , then G has a subgroup of order p^a . We state this without proof. For A_4 it gives a subgroup of order 4, which is the Klein four-group of Example 5.7.

The polyhedral groups of §4.6 can now be named.

Proposition 5.19. *The rotation group of the regular tetrahedron is isomorphic to A_4 , and that of the cube to S_4 .*

Proof. A rotation of the tetrahedron permutes its four vertices. Their position vectors measured from the centre span $\mathbb{R}^3$, so a rotation fixing all four is the identity and the resulting homomorphism into S_4 is injective. The identity, the eight rotations through $\pm 2\pi/3$ about the four vertex axes, and the three half-turns about the axes through midpoints of opposite edges account for all 12 rotations,

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by Theorem 4.35. They give the identity, 3-cycles and double transpositions, all even, so the image is a subgroup of A_4 with 12 elements and equals A_4 .

A rotation of the cube permutes its four body diagonals. The six half-turns about axes through midpoints of opposite edges give the six transpositions, which generate S_4 , so the homomorphism is onto. The rotation group has 24 elements, as does S_4 , so it is an isomorphism. $\square$

We state without proof that the rotation group of the icosahedron is isomorphic to A_5 . Five cubes can be inscribed in the dodecahedron, each using one diagonal of every face, and the rotations permute them.

Quiz 5.3. Is the set of odd permutations a subgroup of S_n ? Check that

$$\text{sgn}((1\ 2\ 3)(4\ 5)) = -1$$

in two ways, by counting transpositions and by Proposition 5.13.

5.4. CONJUGACY CLASSES OF S_n AND YOUNG DIAGRAMS

READING. [Moore, §7.5, pp. 106–108, §7.5.2, pp. 122–124].

Proposition 5.20. For $\sigma \in S_n$ and a cycle $(a_1 \ \cdots \ a_r)$,

$$\sigma(a_1\ a_2\ \cdots\ a_r)\sigma^{-1} = (\sigma(a_1)\ \sigma(a_2)\ \cdots\ \sigma(a_r)).$$

Proof. Apply the left-hand side to $\sigma(a_i)$. First σ^{-1} gives a_i , then the cycle gives a_{i+1} , then σ gives $\sigma(a_{i+1})$, with a_{r+1} read as a_1 . A point outside $\{\sigma(a_1), \dots, \sigma(a_r)\}$ comes from a point outside the cycle and is left alone. The right-hand side does the same. $\square$

Conjugation therefore relabels the entries of a cycle without changing its length, and applying it to a decomposition relabels every cycle at once.

cycle type
轮换型

Definition 5.21 (Cycle type). The *cycle type* of $\varphi \in S_n$ is the tuple $\lambda = (\lambda_1, \dots, \lambda_t)$, where λ_i is the length of the i -th cycle in its complete factorization, ordered so that $\lambda_1 \geq \dots \geq \lambda_t \geq 1$. The lengths sum to n , so λ is a *partition* of n . Write $m_i(\lambda)$ for the number of cycles of length i .

partition
分拆

For example, $(1\ 2)(3\ 4)(5) \in S_5$ has $\lambda = (2, 2, 1)$, with $m_2(\lambda) = 2$ and $m_1(\lambda) = 1$.

Theorem 5.22 (Classes of S_n). Two permutations are conjugate in S_n exactly when they have the same cycle type, so the conjugacy classes of S_n are in bijection with the partitions of n . The class of type λ has size

$$\frac{n!}{\prod_i i^{m_i(\lambda)} m_i(\lambda)!}.$$

Proof. Proposition 5.20 shows that conjugate permutations have the same cycle type. Conversely, let φ and ψ have the same type. Write their complete factorizations with cycles of equal length aligned, and let σ send each entry of φ to the entry standing below it in ψ . This σ is a permutation, and $\sigma\varphi\sigma^{-1} = \psi$ by Proposition 5.20.

For the class size, take the group $G = S_n$ acting on the set $X = S_n$ by conjugation, with the action

$$\sigma \cdot \varphi = \sigma\varphi\sigma^{-1}.$$

The orbit of φ is its conjugacy class $\text{Cl}(\varphi)$. Its stabilizer consists of the permutations whose conjugation leaves φ unchanged. This is the centralizer

$$G_\varphi = \{\sigma \in S_n : \sigma\varphi\sigma^{-1} = \varphi\} = C_{S_n}(\varphi).$$

By Theorem 4.20, $|\text{Cl}(\varphi)| = n!/|C_{S_n}(\varphi)|$. To count this stabilizer, note that a stabilizing σ leaves every power of φ unchanged under conjugation, so $\sigma(\varphi^k(x)) = \varphi^k(\sigma(x))$. Thus σ maps each cycle onto a cycle of the same length, and the image of one point determines the images of all points in that cycle. For the $m_i = m_i(\lambda)$ cycles of length i , there are $m_i!$ ways to assign target cycles and i choices for the image of a chosen point in each cycle, giving i^{m_i} choices. Conversely, each such assignment defines a permutation leaving φ unchanged under conjugation. The choices for different lengths are independent, giving

$$|C_{S_n}(\varphi)| = \prod_i i^{m_i(\lambda)} m_i(\lambda)!.$$

Substituting into the orbit–stabilizer formula gives the class size. $\square$

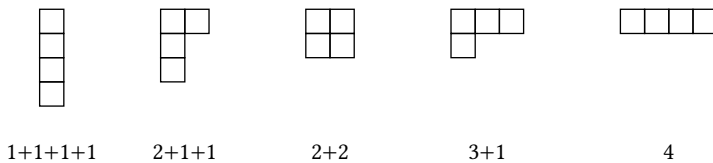
Example 5.23 (The classes of S_4). There are $p(4) = 5$ partitions of 4, hence five classes.

partition	representative	class size	order
$1 + 1 + 1 + 1$	e	1	1
$2 + 1 + 1$	$(1\ 2)$	6	2
$2 + 2$	$(1\ 2)(3\ 4)$	3	2
$3 + 1$	$(1\ 2\ 3)$	8	3
4	$(1\ 2\ 3\ 4)$	6	4

The sizes add to $1 + 6 + 3 + 8 + 6 = 24 = |S_4|$, as the classes partition the group, and each divides 24 as (4.1) requires.

Young diagram
杨图

A partition is drawn as a *Young diagram*, a left-justified array of boxes with λ_i boxes in the i -th row. The five partitions of 4 are



partition function
分拆函数

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and they label the five classes of Example 5.23 in the same order. The number of classes of S_n is the number of partitions of n , the *partition function* $p(n)$, which runs 1, 2, 3, 5, 7, 11, 15, ... for $n = 1, 2, \dots$ and has no simple closed form.

Young tableau
杨表

Definition 5.24 (Young tableau). A *Young tableau* is a Young diagram with n boxes filled with $1, \dots, n$, each used once. It is *standard* when entries increase from left to right in every row and from top to bottom in every column.

For shape $(2, 1)$ there are $3! = 6$ tableaux, of which exactly two are standard.

1	2
3	

1	3
2	

Remark 5.25 (One shape, two uses). Here a Young diagram records cycle lengths and labels a conjugacy class. In Part II the same shapes will also label irreducible representations through constructions that symmetrise tensor positions along rows and antisymmetrise them along columns.

Remark 5.26 (Conjugacy classes in A_4). The eight 3-cycles form a single class of S_4 . In A_4 the centralizer of $(1\ 2\ 3)$ is $\langle (1\ 2\ 3) \rangle$, of order 3, so by (4.1) its A_4 -class has $12/3 = 4$ elements. The eight 3-cycles split into two classes of four, and the conjugating element that merges them, a transposition, is missing from A_4 .

Quiz 5.4. List the partitions of 5 and the class sizes of S_5 , and check that they sum to 120. Which classes of S_5 lie in A_5 ? *Challenge.* Do any of them split into more than one A_5 -class?

5.5. PERMUTATIONS IN PHYSICAL SYSTEMS

Permutation symmetry occurs in coupled oscillators, tensor indices, and identical-particle states.

5.5.1. Identical coupled oscillators

Example 5.27 (A collective motion and relative motions). Consider $n \geq 2$ identical scalar coordinates q_i , with mass $m > 0$, the same restoring constant $k_0 > 0$, and the same coupling $k > 0$ between every pair. Their kinetic and potential energies are

$$T = \frac{m}{2} \sum_{i=1}^n \dot{q}_i^2, \quad U = \frac{k_0}{2} \sum_{i=1}^n q_i^2 + \frac{k}{2} \sum_{i < j} (q_i - q_j)^2.$$

A permutation acts by $(\sigma \cdot q)_i = q_{\sigma^{-1}(i)}$, the matrix action of Definition 5.11. Both energies are unchanged. The first sums run over every coordinate, and the second over every unordered pair.

Separate the mean displacement from the relative displacements,

$$Q = \frac{1}{n} \sum_i q_i, \quad r_i = q_i - Q, \quad \sum_i r_i = 0.$$

Relabelling leaves the collective pattern $(Q, \dots, Q)$ unchanged and preserves $\sum_i r_i = 0$. The collective displacement has one independent amplitude Q . The relative displacements have $n-1$ independent amplitudes, since their n components obey one sum constraint. The equations of motion are

$$m\ddot{q}_i = -k_0 q_i - k \sum_{j \neq i} (q_i - q_j) = -k_0 q_i - nk(q_i - Q).$$

Taking the mean, then subtracting it from each equation, gives

$$m\ddot{Q} = -k_0 Q, \quad m\ddot{r}_i = -(k_0 + nk)r_i.$$

There is one collective mode with $\omega^2 = k_0/m$, and $n-1$ independent relative modes with the same $\omega^2 = (k_0 + nk)/m$.

More generally, write $U = \frac{1}{2} q^T D q$, where D is the symmetric stiffness matrix. Invariance under relabelling means

$$P(\sigma)^T D P(\sigma) = D, \quad D_{\sigma(i), \sigma(j)} = D_{ij}.$$

Relabelling can send any coordinate to any other coordinate, and any pair of distinct coordinates to any other such pair. Thus all diagonal entries of D are equal, and all off-diagonal entries are equal. Writing these entries as $a + b$ and b , respectively, gives

$$D = a\mathbf{1} + buu^T, \quad u = (1, \dots, 1)^T.$$

For a collective displacement $q = Qu$, all coordinates move together. Since $u^T u = n$,

$$D(Qu) = (a + nb)Qu.$$

Thus the uniform pattern u is an eigenvector of D with eigenvalue $a + nb$. For a relative displacement $r = (r_1, \dots, r_n)^T$, the condition $\sum_i r_i = 0$ gives $u^T r = 0$, so

$$Dr = ar.$$

Every relative pattern is therefore multiplied by the same eigenvalue a . Since the equations of motion are $m\ddot{q} = -Dq$, the collective mode has $\omega^2 = (a + nb)/m$, and all $n-1$ relative modes have $\omega^2 = a/m$. In this example, $a = k_0 + nk$ and $b = -k$, recovering the two frequencies above.

5. The symmetric group

5.5.2. Exchanging indices and rotating components

We can exchange the two index positions of an array A_{ij} or rotate both indices. For a real 3×3 array these operations are

$$\tau(A) = A^T, \quad \mathcal{R}_R(A) = RAR^T, \quad R \in SO(3).$$

Here $\tau^2 = \text{id}$, so τ defines an S_2 action on the *two index positions* in A_{ij} , with each index ranging over 1, 2, 3. The rotations give a second action, and the two commute, since

$$(RAR^T)^T = RA^T R^T.$$

Every array has the unique decomposition

$$A = S + K, \quad S = \frac{A + A^T}{2}, \quad K = \frac{A - A^T}{2}.$$

Thus $\tau(S) = S$ and $\tau(K) = -K$. The two operations $\Pi_{\pm} = (\text{id} \pm \tau)/2$ satisfy $\Pi_{\pm}^2 = \Pi_{\pm}$ and $\Pi_+ \Pi_- = 0$. They project onto the symmetric and antisymmetric arrays. These are the simplest examples of the symmetrising and antisymmetrising operations later associated with a row of two boxes and a column of two boxes. Because rotations commute with τ , they preserve both spaces.

For a 3×3 array, this S_2 action separates the nine components into six symmetric and three antisymmetric components. Rotations further separate the symmetric part into its trace and a traceless part,

$$S = \frac{\text{Tr } S}{3} \mathbf{1} + S_0, \quad \text{Tr } S_0 = 0.$$

The trace supplies one independent number, while the traceless condition leaves five independent symmetric components. Since $R \mathbf{1} R^T = \mathbf{1}$ and $\text{Tr}(RSR^T) = \text{Tr } S$, rotations preserve these parts separately. Thus

$$9 = (1 + 5) + 3.$$

The inertia tensor provides a symmetric array; a quadrupole moment is described by a symmetric traceless one. An antisymmetric array can be written as $K_{ij} = \sum_k \epsilon_{ijk} a_k$ and carries a vector under proper rotations. In Part II the 1, 3, and 5 dimensional spaces will be identified with angular momenta 0, 1, and 2.

For an array with n index positions, the index exchange generalises to an S_n action, which commutes with applying the same matrix g to every index. These commuting actions connect Young diagrams with representations of matrix groups, as developed in Part II.

5.5.3. Particle exchange

The exchange of two particles in §1.3.2 extends to n particles. Let P_{σ} be the operator permuting their labels, with $P_{\sigma} P_{\rho} = P_{\sigma\rho}$. The usual bosonic and fermionic exchange rules are respectively

$$P_{\sigma}\psi = \psi, \quad P_{\sigma}\psi = \text{sgn}(\sigma)\psi.$$

Since sgn is a homomorphism, the fermionic rule gives the same sign for every sequence of pair exchanges producing σ . All one-particle variables, including spin, are exchanged together, so these rules apply to the full state.

KEY POINTS

- Disjoint cycles are the orbits of $\langle \varphi \rangle$; their lengths determine the order of φ by taking the least common multiple.
- Cayley's theorem embeds G in S_G through left multiplication.
- $\text{sgn}(\varphi) = \det P(\varphi)$ makes parity well defined. Its kernel A_n has order $n!/2$ for $n \geq 2$.
- The classes of S_n are its cycle types, labelled by partitions of n and drawn as Young diagrams.
- Permutations organise physical coordinates and tensor positions. Rotations preserve index symmetry but may split these spaces further.

KEY TERMS. cycle 轮换 · transposition 对换 · cycle decomposition 轮换分解 · Cayley's theorem 凯莱定理 · sign 符号 · even permutation 偶置换 · odd permutation 奇置换 · alternating group 交错群 · cycle type 轮换型 · partition 分拆 · Young diagram 杨图 · Young tableau 杨表